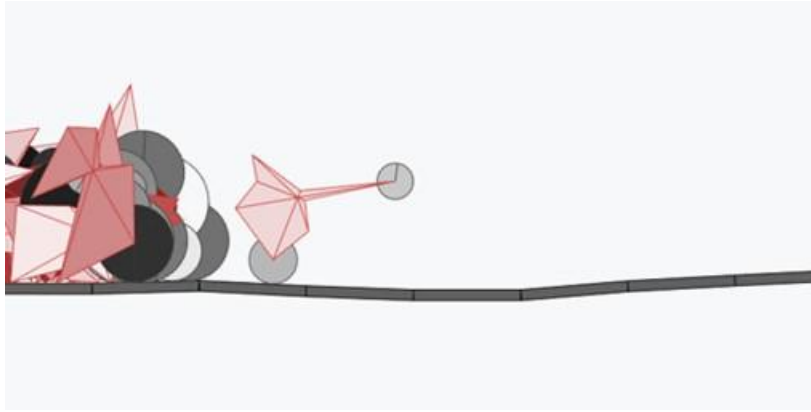


Lab report, part B, number 2 - 2025.

Check out the evolutionary algorithm simulator at

https://rednuht.org/genetic_cars_2/



The objective of this lab is to try out a few different learning parameters on this sim to get an understanding of how evolutionary algorithms work. The sim uses a number of different weight and shape parameters for cars/bikes, and fitness is measured by how far they are able to drive on a track of gradually increasing difficulty. The Genome of the simulation is listed down near the bottom of the page.

Step 1: Enter the new world creation seed of CSC357, and hit GO. We all want to use the same seed to more easily compare results. Observe the different Cars/bikes and see which properties seem to result in the farthest distances. Run for as long as you want till you think you have a decent winning result.

Step2: try two additional combinations of different mutation rate, mutation size and elite clones.

Write up submission for your report: (Max ¾ page in total for this one)

- Describe which aspects of the genome seemed to have the largest impact on fitness
- Did you find much variation in successful individuals? Meaning, was there more than one way of going far?
- what impact did the change in learning parameters have on your winning distances? Why do you think?